Online Event Management System

https://github.com/aiyshwary/Online-Event-Management-System

SQL NoSQL Database Design Normalization

OVERVIEW

A database-driven system for managing events end-to-end, allowing customers to register, browse and book events with discounts, while giving administrators full control over event listings, users, payments and preferences.

IMPACT

Designed and implemented a normalized relational database and NoSQL schema for an Online Event Management System, covering customer registration, booking workflows, discount management, and payment processing.

TECHNICAL WALKTHROUGH

Online Event Management System – Project Explanation This project is a database-driven Online Event Management System designed to help users register, browse, and book events online, while allowing admins to manage events, users, discounts, and payments efficiently.

Problem it Solves Managing events manually is time-consuming and error-prone—especially when handling: Multiple event types User registrations Discounts and special preferences Payments and bookings This system provides a centralized, structured database that automates all these processes.

Key Actors

User

Registers using email and password Browses events (Dance, Music, Arts, Sports, etc.) Books events with number of seats Applies discounts and special preferences Makes payment using multiple modes Can update or cancel bookings

Admin

Logs in with admin credentials Adds, updates, or deletes: Events Discounts Special preferences User details Manages pricing and discount rates Core Database Modules (Entities) User Stores registered user details: User ID, Email, Username, Password Users must register before booking any event.

Event Details Stores complete event information: Event name, type, date & time Duration and base price Each event belongs to a specific event category.

Event Discount

Manages discount options: Premium account Group booking Coupon-based discounts Each discount has a rate reduced value.

Special Preference Optional add-ons for users: VIP seats Backstage access Exclusive services These come with extra charges.

User Booking Acts as the core linking table: Connects User, Event, Discount, and Special Preference Stores number of seats booked Each booking has a unique Booking ID (BID) This table enforces referential integrity using foreign keys.

Amount (Payment) Handles payment details: Payment ID Booking ID Mode of payment (Credit, Debit, Wallet, etc.) Admin Stores admin login credentials and permissions.

Database Design Highlights Fully normalized (3NF) to avoid redundancy Strong foreign key relationships Uses indexes for faster search Supports views for business insights Important SQL Features Used Queries Simple queries (filter, sort) Aggregate queries (COUNT, SUM, AVG) Join queries (INNER, LEFT, RIGHT joins)

Views

Examples: Total amount payable by each user (including discounts & extra charges) Users who registered but never booked Number of bookings per user Events sorted by price for easy searching Indexes Faster event search by type & name Unique email constraint for users Faster lookup for bookings, discounts, and prices Overall Summary (One-Line) This project implements a relational database–based Online Event Management System that efficiently handles user registrations, event bookings, discounts, special preferences, and payments with strong data integrity and optimized query performance.

KEY DETAILS

- Designed a fully normalized relational schema (up to 3NF) with entities for Users, Events, Bookings, Payments, and Discounts.
- Modeled equivalent NoSQL document structures for flexible event-type storage and preference handling.
- Admin interface supports CRUD operations on events, user management, and discount configuration.
- Customer portal enables account creation, event browsing, booking with preference selection, and payment tracking.
- Schema documented with ER diagrams, normalization steps, and a detailed project report.